

MODERN AI/ML P

Prediction Before Explanation • Failure Mo

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#	Phase	Month	Section
1	Phase 1	Month 1	The Reasoning Layer
2	Phase 1	Month 1	The Reasoning Layer
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5	Phase 1	Month 1	The Reasoning Layer
6	Phase 1	Month 1	The Reasoning Layer
7	Phase 1	Month 2	Classical ML
8	Phase 1	Month 2	Classical ML
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12	Phase 1	Month 2	Classical ML
13	Phase 1	Month 2.5	Intro MLOps & Deployment
14	Phase 1	Month 2.5	Intro MLOps & Deployment
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16	Phase 1	Month 2.5	Intro MLOps & Deployment
17	Phase 1	Month 3	Deep Learning Foundations
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21	Phase 1	Month 3	Deep Learning Foundations
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23	Phase 2	Month 4	GenAI & Embeddings
24	Phase 2	Month 4	GenAI & Embeddings
25	Phase 2	Month 4	GenAI & Embeddings
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27	Phase 2	Month 4	GenAI & Embeddings
28	Phase 2	Month 4	GenAI & Embeddings
29	Phase 2	Month 4	GenAI & Embeddings
30	Phase 2	Month 4.5	Advanced MLOps
31	Phase 2	Month 4.5	Advanced MLOps
32	Phase 2	Month 4.5	Advanced MLOps
33	Phase 2	Month 5	System Design & Production ML
34	Phase 2	Month 5	System Design & Production ML
35	Phase 2	Month 5	System Design & Production ML
36	Phase 2	Month 5	System Design & Production ML
37	Phase 2	Month 5	System Design & Production ML
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42	Phase 3	Month 6-8	Advanced Production
43	Phase 3	Month 6-8	Advanced Production
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45	Phase 3	Month 6-8	Advanced Production
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54	Phase 3	Month 6-8	Advanced Production

ROADMAP — FIVE LAWS OF LEARNING >

Codes Over Features • Compression Beats Coverage • Emotion Creates Judgment

10 sheets: Phase 1 Foundation (22 topics) | Phase 2 Compression (19 topics) | Phase 3 Ownership (13 topics)

Topic
Python Fundamentals
NumPy
Pandas
Statistics
Linear Algebra
Calculus
Linear Regression (Deep Dive — Compression Anchor)
Logistic Regression
Decision Trees
Random Forests
Gradient Boosting (XGBoost, LightGBM, CatBoost)

Unsupervised Learning (Clustering, PCA, Anomaly Detection)
Git & Version Control
Docker Fundamentals
Basic Model Deployment (FastAPI)
Environment Management
Neural Networks from Scratch
PyTorch Fundamentals
Optimization Algorithms
CNNs (Convolutional Neural Networks)
RNNs (Recurrent Neural Networks)
Transformers
Embeddings
Vector Databases
LLMs (Large Language Models)
Prompt Engineering

Cost Mathematics for LLMs
RAG (Retrieval-Augmented Generation)
Fine-tuning LLMs
Experiment Tracking
Model Registry
Data Versioning
System Design for ML
Hybrid Architecture Design
Caching Strategies
Fallback Chains
Load Balancing & Scaling
API Design for ML Systems
Monitoring & Observability
Model Drift Detection
Logging & Debugging

CI/CD for ML
A/B Testing Frameworks
Model Evaluation in Production
Cost Optimization Advanced
Feature Engineering in Production
Data Preprocessing Pipelines
Model Interpretability
Incident Response & Post-mortems
Security for ML Systems
Infrastructure as Code
Batch Processing & Pipelines
Streaming Data Processing
Multi-model Serving

✕ 54 TOPICS

- AI Accelerates, Humans Judge

3 topics)

Key Subtopics

Reading code (not just writing) • Code comprehension & debugging • Best practices & organization

Array thinking vs iteration • Vectorization • Broadcasting • Memory efficiency • Universal functions

DataFrame operations • Data manipulation • Performance optimization • Real-world data cleaning • GroupBy & merge

Quantifying uncertainty • Hypothesis testing • Probability distributions • Sampling • Confidence intervals • Bayesian thinking

Geometric intuition • Matrix operations • Vector spaces • Eigenvalues/eigenvectors • SVD • ML applications

Derivatives & partial derivatives • Chain rule (backprop foundation) • Gradient intuition • Optimization fundamentals • Integrals for probability

Mathematical foundations • Assumptions & violations • Failure modes • Cost functions • Regularization (Ridge/Lasso) • Residual analysis

Link functions • Binary/multi-class • Decision boundaries • Sigmoid derivation • Threshold selection • Calibration

Splitting criteria (Gini, Entropy) • Pruning • Overfitting prevention • Interpretability • Feature importance

Ensemble methods • Bagging • Feature importance • Out-of-bag error • Feature subsampling

Sequential trees • XGBoost • LightGBM • CatBoost • Silent failures • Hyperparameter tuning

K-Means • DBSCAN • Hierarchical clustering • PCA • t-SNE/UMAP • Anomaly detection • Silhouette score
Repository management • Branching • Collaborative workflows • Code review • .gitignore for ML
Why containerize • Dockerfile creation • Image optimization • Container basics • Docker Compose
FastAPI basics • REST APIs • Model serving patterns • Health checks • Request/response schemas
Virtual environments • Dependency management (pip, poetry) • Requirements files • Reproducibility
Backpropagation (NumPy) • Gradient computation • Activation functions • Loss functions • Weight initialization
Autograd & computational graphs • Tensors • nn.Module • Training loops • Saving/loading • GPU management
SGD • Momentum • Adam • Learning rate scheduling • When to use which • Convergence analysis
Why they work for images • Conv layers • Pooling • Architectures (ResNet, VGG) • Transfer learning
Sequential processing • Vanishing/exploding gradients • LSTM/GRU • When to use vs Transformers • Sequence modeling
Self-attention • Positional encoding • Multi-head attention • Encoder-decoder • Attention visualization
Word embeddings (Word2Vec, GloVe) • Sentence embeddings • Contextual (BERT) • Embedding geometry • Dimensionality reduction
Similarity search algorithms • HNSW • IVF • Flat search • Performance tradeoffs • Pinecone, Chroma, Qdrant
Pre-training • Fine-tuning • RLHF • Architectures (GPT, BERT, T5, LLaMA) • Model families
Chain of thought • Few-shot learning • System prompts • Prompt optimization • Templates • Output formatting

Token pricing • Request cost estimation • Scaling calculations • Business model viability
Architecture • Chunking strategies • Dense vs sparse retrieval • BM25 • Hybrid search • Re-ranking • Eval metrics
When to fine-tune vs prompt • Cost-benefit • Data requirements • LoRA • Evaluation strategies
MLflow • Weights & Biases • Parameter logging • Metric tracking • Model versioning • Comparison dashboards
Versioning strategies • Metadata management • Lineage tracking • Promotion to production • Rollback
DVC • Dataset tracking • Reproducibility • Data lineage • Storage backends
Scalability • Latency optimization • Data pipelines • Streaming vs batch • Fault tolerance • Trade-offs
Rules systems • Classic ML vs LLMs • Tiered routing • Cost optimization • Request routing logic
Exact match • Semantic caching • Fuzzy caching • Cache hit rate • Invalidation strategies
Error handling • Model cascading • Graceful degradation • Circuit breaker • Timeout strategies
Horizontal vs vertical scaling • Load balancing • Auto-scaling • Traffic management • Resource optimization
RESTful principles • Schemas • Rate limiting • Auth • API versioning • Batch endpoints
Prometheus • Grafana • Latency (p50/p95/p99) • Error rates • Cost tracking • Custom metrics
Prediction monitoring • Feature drift • Concept drift • KL divergence • Statistical tests
Structured logging • Log aggregation • Distributed tracing • Production debugging • Profiling (cProfile, PyTorch Profiler)

Continuous integration • Automated ML testing • Continuous deployment • Blue-green • Canary releases
Experiment design • Statistical significance • Traffic splitting • Metrics selection • Interpretation
Business vs technical metrics • Eval frameworks • Online vs offline • Human-in-the-loop • Continuous eval
Model distillation • Quantization (INT8/INT4) • Batching • Dynamic model selection • Routing optimization
Feature stores • Online vs offline features • Feature monitoring • Pipeline optimization • Validation
Missing data • Outlier detection • Data validation • Data augmentation • Pipeline orchestration
SHAP • LIME • Feature importance • Partial dependence • Explaining to stakeholders
On-call practices • Root cause analysis • Post-mortem documentation • Runbook creation • Prevention
Model security • API security • Data privacy • Adversarial attacks • Compliance
Terraform basics • Cloud resource management • Infrastructure versioning • Automated provisioning
Batch inference • Memory management • Throughput vs latency • Orchestration (Airflow, Prefect)
Kafka fundamentals • Stream processing patterns • Real-time predictions • Handling late data
Model ensembling • Routing strategies • Shadow deployments • Graceful transitions